PROJECT PROPOSAL

LLM-Based Long-Term Conversation Memory Optimization

A Research Proposal for Efficient Long-Term Memory in LLM-Based Conversational Agents

Submitted By

Saima Sultana
Student ID: 232-115-231

Jannat Akther Chaity
Student ID: 232-115-215

Department of Computer Science and Engineering
Metropolitan University

Academic Year: 2026

October 1, 2026

Contents

1 Introduction **1**

2 Problem Statement **2**

3 Aim and Objectives **3**

3.1 Aim 3

3.2 Objectives 3

4 Research Questions **4**

5 Literature Review and Research Gap **5**

5.1 Existing Approaches 5

5.2 Research Gap 5

6 Proposed Methodology **7**

6.1 Proposed Architecture 7

6.2 Memory Management 8

6.3 Memory Retrieval 8

7 Dataset and Evaluation **9**

7.1 Datasets and Benchmarks 9

7.2 Baseline Methods 9

7.3 Evaluation Metrics 10

8 System Implementation **11**

9 Expected Outcomes **13**

10 Scope and Limitations **14**

10.1 Scope 14

10.2 Limitations 14

11 Work Plan **15**

12 Conclusion **16**

13 References **17**

1. Introduction

Large Language Models (LLMs) have significantly improved conversational artificial intelligence by enabling systems to understand and generate natural language. However, most conversational systems still face difficulties when interactions continue over long periods. A system may need to remember information from previous conversations, such as a user's name, preferences, goals, or ongoing projects.

A simple solution is to keep the complete conversation history and provide it to the LLM whenever a new message is received. Although this approach preserves information, it can increase context size, token usage, processing cost, and response latency. It may also provide the model with a large amount of irrelevant information.

Another approach is to summarize previous conversations. Summarization reduces the context size but may remove details that become useful later. Retrieval-based systems can retrieve relevant information, but they may still store unnecessary information or retrieve irrelevant memories.

Therefore, an effective long-term memory system should decide:

What should be remembered, what should be forgotten, and what should be retrieved?

This project proposes an optimized long-term memory mechanism for LLM-based conversational agents. The proposed system will selectively store useful information, discard unnecessary information, and retrieve relevant memories when required.

2. Problem Statement

Long-term conversation creates a memory-management problem for LLM-based systems. Not all information in a conversation has the same future value.

For example:

Table 2.1: Example of Memory Selection

Information	**Decision**
My name is Saima.	Store
I prefer Python examples.	Store
I am working on an LLM project.	Store
I am drinking coffee right now.	Discard
The weather is nice today.	Discard

If all information is stored, memory can become unnecessarily large. If too much information is removed, important knowledge may be lost.

The main problem addressed by this research is therefore:

> **How can an LLM-based conversational system selectively store useful information and discard unnecessary information while maintaining high-quality long-term conversations?**

3. Aim and Objectives

3.1 Aim

The main aim of this project is to develop and evaluate an optimized long-term memory mechanism for LLM-based conversational agents that can selectively retain useful information and retrieve relevant memories while reducing unnecessary memory and context usage.

3.2 Objectives

The project will:

1. Study existing long-term memory techniques for LLM-based conversational systems.
2. Identify characteristics that make conversational information useful for future interactions.
3. Develop a mechanism for deciding whether information should be stored, updated, or discarded.
4. Store useful information as structured long-term memory.
5. Retrieve relevant memories when responding to future queries.
6. Compare the proposed approach with full-history, summarization, and retrieval-based approaches.
7. Evaluate memory quality, response performance, token usage, memory size, and latency.
8. Implement the proposed mechanism in a MERN-based conversational application.

4. Research Questions

The main research question is:

> **Can long-term conversation memory be optimized by selectively storing useful information and discarding unnecessary information while maintaining response quality?**

The following questions will support the research:

- **RQ1:** How can useful and unnecessary conversational information be identified?
- **RQ2:** How does selective memory affect long-term conversational performance?
- **RQ3:** How much memory and context usage can be reduced through selective storage?
- **RQ4:** How does the proposed method compare with full-history, summarization, and retrieval-based approaches?
- **RQ5:** What relationship exists between memory size, retrieval quality, context usage, and response quality?

5. Literature Review and Research Gap

Long-term memory for LLM-based agents is an active area of research. Existing studies have investigated long-context processing, conversation summarization, retrieval-augmented generation, external memory, and memory mechanisms for autonomous agents.

5.1 Existing Approaches

Full Conversation History: The complete conversation is retained and provided to the model. This preserves historical information but can result in high context and token usage.

Conversation Summarization: Previous conversations are compressed into summaries. This reduces context size but may remove details that become important later.

Retrieval-Based Memory: Conversation information is stored externally and relevant memories are retrieved according to the current query. This can reduce context usage, but retrieval quality becomes important.

Selective Memory: The system evaluates information before storing it and attempts to retain only information that may be useful in future conversations.

5.2 Research Gap

The proposed research focuses on the combination of memory selection and memory retrieval. Instead of storing every conversational detail, the system will evaluate information based on factors such as:

- Importance

- Future usefulness
- Relevance
- Recurrence
- Temporariness
- Redundancy

The research will investigate whether selective memory can reduce unnecessary memory and context usage while preserving useful information for future conversations.

A literature review of approximately 15–20 papers will be conducted, with around 10–12 highly relevant papers selected for detailed analysis. Sources will include Google Scholar, Semantic Scholar, IEEE Xplore, ACM Digital Library, and arXiv.

Search terms will include:

- LLM long-term memory
- conversational memory
- LLM agent memory
- memory retrieval
- selective memory
- memory compression
- LongMemEval
- LoCoMo

6. Proposed Methodology

The proposed methodology consists of memory evaluation, storage, retrieval, and response generation.

6.1 Proposed Architecture

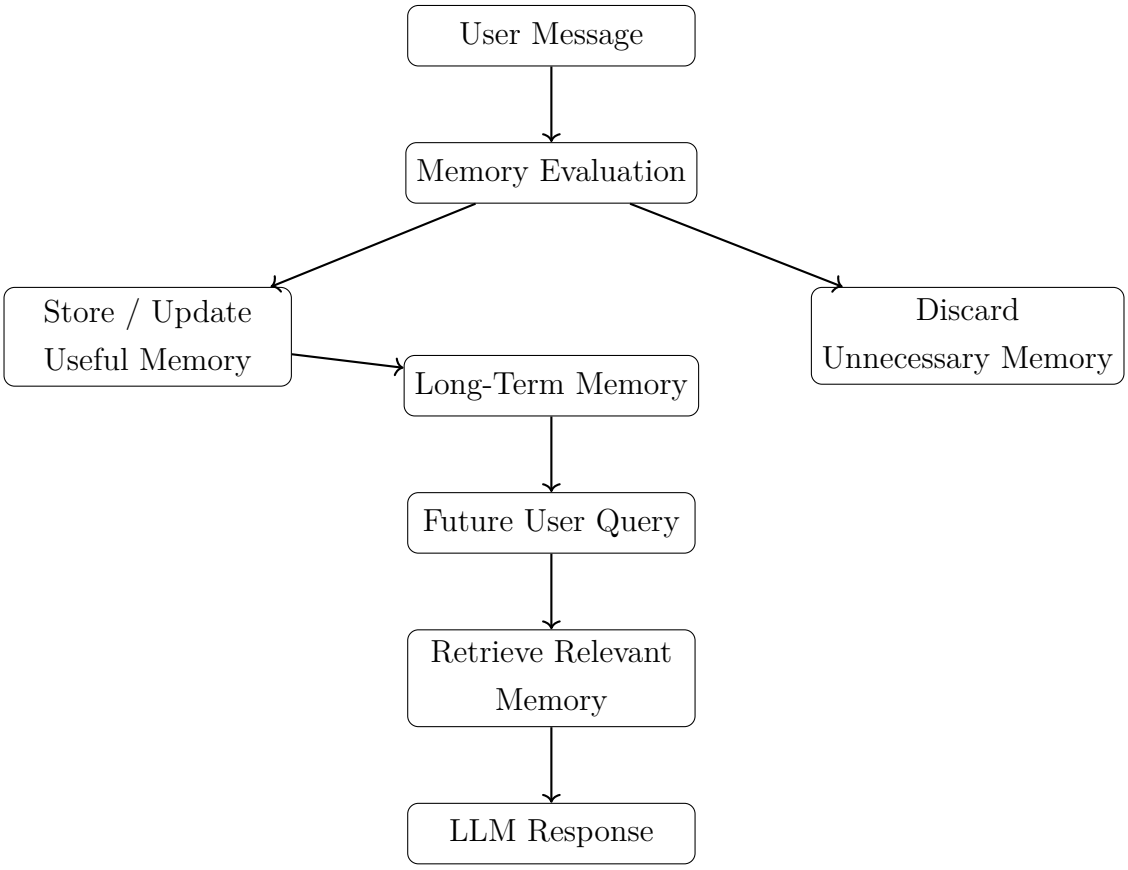


Figure 6.1: Proposed Long-Term Memory Architecture

6.2 Memory Management

When a user sends a message, the system will evaluate whether the information should become long-term memory.

Possible actions are:

STORE **UPDATE** **KEEP** **REPLACE** **DISCARD**

Stored memories may contain:

- Memory content
- Memory type
- Importance score
- Timestamp
- Embedding
- Source conversation
- Retrieval frequency

Memory categories may include personal information, preferences, goals, tasks, and long-term project information.

6.3 Memory Retrieval

When a future query arrives, the system will retrieve memories that are relevant to the query.

For example, if the stored memory is:

> "The user prefers Python examples with explanations."

and the user later asks:

> "How should you teach me programming?"

the system should retrieve the relevant preference and provide it to the LLM.

The basic process is:

$$\text{Current Query} + \text{Relevant Memories} \rightarrow \text{LLM} \rightarrow \text{Response}$$

7. Dataset and Evaluation

7.1 Datasets and Benchmarks

The research will primarily use existing conversational-memory benchmarks rather than creating a large new dataset.

LongMemEval will be considered as the primary benchmark for evaluating long-term memory capabilities.

LoCoMo and other suitable benchmarks may also be considered depending on the final literature review and experimental requirements.

7.2 Baseline Methods

The proposed method will be compared with:

1. Full conversation history
2. Conversation summarization
3. Retrieval-based memory
4. Proposed selective memory

Table 7.1: Planned Evaluation Comparison

Method	**Memory Strategy**	**Main Evaluation**
Full History	Complete conversation	Context and response quality
Summarization	Compressed conversation	Context reduction
Retrieval	Query-based retrieval	Retrieval quality
Proposed	Selective storage + retrieval	Quality and efficiency

7.3 Evaluation Metrics

The system will be evaluated using three groups of metrics.

Conversational Performance

- Answer accuracy
- Long-term question answering
- Response quality
- Memory retrieval accuracy

Memory Quality

- Useful memory retention
- Unnecessary-memory rate
- Irrelevant retrieval
- Redundant memory

Efficiency

- Number of stored memories
- Number of retrieved memories
- Context/token usage
- Token reduction
- Response latency

The objective is to determine whether memory can be reduced without significantly reducing useful conversational information.

8. System Implementation

A MERN-based application will be developed to demonstrate the proposed memory mechanism.

The system will contain four main components:

- **React:** User interface and chat interface.
- **Node.js and Express:** Backend API and application logic.
- **MongoDB:** Storage for conversations, memories, and metadata.
- **LLM and Retrieval Module:** Response generation and relevant-memory retrieval.

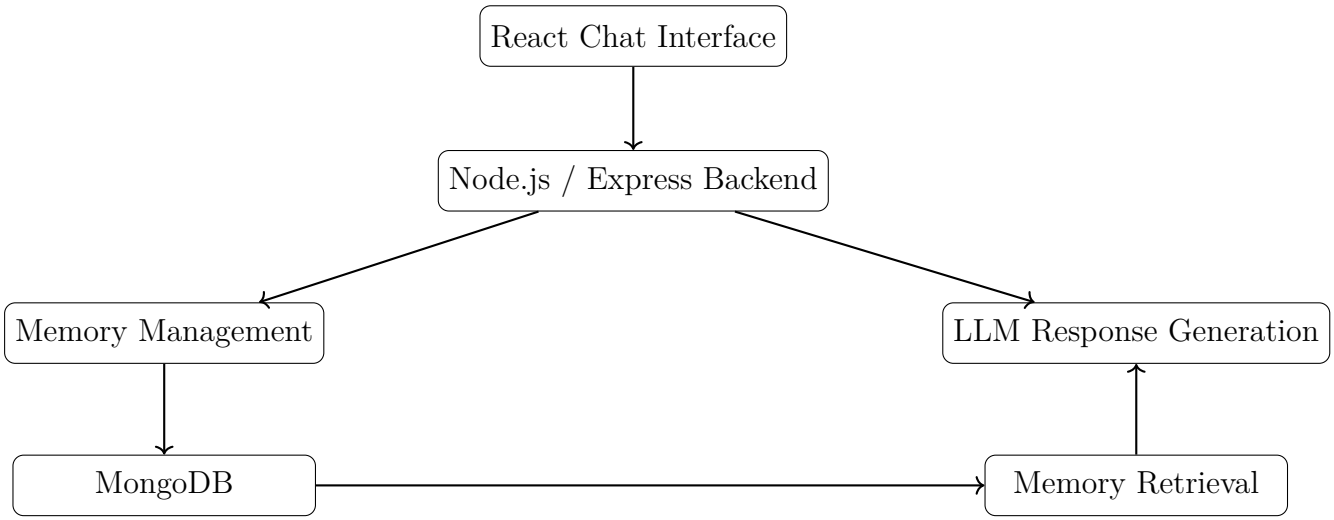


Figure 8.1: MERN-Based System Architecture

The application will demonstrate the complete memory lifecycle:

1. Receive user message.
2. Evaluate memory importance.

3. Store, update, or discard information.
4. Retrieve relevant memories for future queries.
5. Provide relevant memory to the LLM.
6. Generate the final response.

9. Expected Outcomes

The project is expected to produce:

1. A selective long-term memory mechanism for LLM-based conversational agents.
2. A method for identifying useful, temporary, redundant, and unnecessary information.
3. A memory storage and retrieval system that provides relevant information to the LLM.
4. Experimental results comparing the proposed approach with existing memory strategies.
5. An analysis of memory size, context usage, response quality, retrieval performance, and latency.
6. A functional MERN-based conversational AI application demonstrating the proposed approach.

The research will determine through experiments whether selective memory can improve memory efficiency while maintaining useful conversational performance.

10. Scope and Limitations

10.1 Scope

The project focuses on long-term memory management for LLM-based conversational systems. It covers:

- Selective memory storage
- Memory updating and removal
- Relevant-memory retrieval
- Memory and context efficiency
- Long-term conversational performance
- MERN-based implementation

The project will not involve training an LLM from scratch.

10.2 Limitations

Potential limitations include:

- Future usefulness of information can be difficult to determine automatically.
- Temporary information may require expiration mechanisms.
- Benchmark datasets may not represent every real-world conversation.
- Retrieval performance may depend on the selected embedding and retrieval model.
- Results may vary depending on the underlying LLM.

11. Work Plan

Table 11.1: Proposed Project Work Plan

Phase	**Activities**	**Duration**
1	Literature review and paper selection	Weeks 1–3
2	Research gap and methodology design	Weeks 4–5
3	Memory system development	Weeks 6–9
4	Dataset and baseline preparation	Weeks 10–11
5	Experiments and evaluation	Weeks 12–15
6	MERN application development	Weeks 16–18
7	Final analysis and documentation	Weeks 19–22

12. Conclusion

Long-term memory is an important requirement for conversational systems that are expected to maintain useful information across multiple interactions. Storing the complete conversation history may increase context and computational requirements, while excessive information removal may result in the loss of useful knowledge.

This project proposes a selective long-term memory mechanism that evaluates conversational information, stores useful memories, removes unnecessary information, and retrieves relevant memories when required.

The proposed approach will be evaluated against full-history, summarization, and retrieval-based methods using suitable conversational-memory benchmarks. The evaluation will consider both conversational performance and efficiency.

Finally, the proposed mechanism will be implemented in a MERN-based conversational application. The project aims to provide both an experimental investigation and a practical demonstration of efficient long-term memory for LLM-based conversational agents.

13. References

The final reference list will be prepared after the detailed literature review. The review will focus on research related to:

1. Long-term memory for Large Language Models.
2. Conversational memory for LLM-based agents.
3. Retrieval-augmented generation and semantic retrieval.
4. Memory compression and context optimization.
5. Selective memory and memory management.
6. LongMemEval and long-term conversational evaluation.
7. LoCoMo and conversational-memory benchmarks.